

Current Sensing, ADC Resolution, and SOC Design

Battery Management System for Electric Scooter | ECE 445 Team 35 | Technical Design Note

Design Objective

The master board measures pack current so that firmware can detect overcurrent conditions and estimate state of charge using coulomb counting. The final implementation uses a low-side shunt resistor and a TI INA240 current-sense amplifier.

- Shunt resistance: 1 mOhm.
- Amplifier gain: 100 V/V.
- Design current range: 5 A minimum measurement case to 30 A maximum measurement case.
- ADC range: STM32 input from 0 V to 3.3 V.

Shunt Voltage Calculation

The shunt converts current into a small differential voltage according to Ohm's law.

$$V_{\text{shunt}} = I_{\text{pack}} * R_{\text{shunt}}$$

$$V_{\text{shunt,max}} = 30 \text{ A} * 0.001 \text{ Ohm} = 30 \text{ mV}$$

$$V_{\text{shunt,min}} = 5 \text{ A} * 0.001 \text{ Ohm} = 5 \text{ mV}$$

These voltages are too small to use directly with good ADC resolution, so the INA240 amplifies them before the MCU ADC samples the signal.

Amplified ADC Input Range

With amplifier gain $G = 100 \text{ V/V}$, the ADC input voltage is:

$$V_{\text{ADC}} = G * V_{\text{shunt}}$$

$$V_{\text{ADC,max}} = 100 * 30 \text{ mV} = 3.0 \text{ V}$$

$$V_{\text{ADC,min}} = 100 * 5 \text{ mV} = 0.5 \text{ V}$$

This maps the intended current range into a useful portion of the 0 V to 3.3 V STM32 ADC input range without saturating the ADC at 30 A.

Pack Current	Shunt Voltage	Amplifier Output
5 A	5 mV	0.5 V
30 A	30 mV	3.0 V

ADC Resolution

For a 12-bit ADC over a 3.3 V range, one least significant bit is approximately 0.8 mV at the ADC input.

$$\text{ADC_LSB} = 3.3 \text{ V} / 4096 = 0.000805 \text{ V} = 0.805 \text{ mV}$$

Referred back to the shunt input through the 100 V/V amplifier:

$$V_{\text{shunt,LSB}} = \text{ADC_LSB} / G = 0.805 \text{ mV} / 100 = 8.05 \text{ uV}$$

$$I_{\text{LSB}} = V_{\text{shunt,LSB}} / R_{\text{shunt}} = 8.05 \text{ uV} / 1 \text{ mOhm} = 8.05 \text{ mA}$$

The resulting current resolution is approximately 8 mA per ADC count, which is sufficiently fine for prototype SOC integration and current reporting.

Coulomb Counting SOC Equation

SOC is estimated by integrating current over time. Positive current is treated as discharge in this convention.

$$\text{SOC}(t) = \text{SOC}(t_0) - (1 / Q_{\text{nom}}) * \int_{t_0}^t I(\tau) d\tau$$

Because current is integrated, offset and gain errors can accumulate into SOC drift. This is why the INA240 was chosen: low offset, low gain drift, and strong PWM rejection improve long-term current integration accuracy in a switching-noise environment.

Measured Current Verification

The current measurement path was checked against a digital multimeter reference. At a representative load, the DMM measured 2.01 A while the UART-reported value was 1.97 A.

$$\text{Percent error} = |2.01 \text{ A} - 1.97 \text{ A}| / 2.01 \text{ A} * 100\% = 1.99\%$$

The measured error is below the +/-5% acceptance criterion for the current-sensing requirement.